string theorists are a little downcast, loop theorists are buoyant, but it is still a matter of clues; there is no strong evidence at all.

To find more concrete confirmation of the theory, we need to look elsewhere. The primordial universe could open the window to predictions capable of confirming the theory. In a not too distant future, we hope. Or they could prove the theory wrong.

A window on to quantum gravity

If we have the equations that describe the transition of the universe across the quantum phase, we can compute effects of quantum phenomena upon the universe which we observe today. The universe is filled with cosmic radiation: a sea of photons remained in the cosmos since the early hot phase, the residual glare of the early high temperature.

The electromagnetic field in the immense space between galaxies trembles like the surface of the sea after a big storm. This quivering, disseminated throughout the universe, is called the cosmic background radiation. It has been studied in the past few years by satellites such as COBE, WMAP and, most recently, Planck. An image of the minute fluctuations of this radiation is given in figure 9.3. The details of the structure of this radiation tell us the history of the universe and, hidden in the folds of these details, there could be footprints of the quantum beginning of our universe.

One of the most active sectors of research in loop quantum gravity is studying how the quantum dynamic of the primordial universe is reflected in this data. The results are preliminary, but encouraging. With more calculations and more precise measurements, it should be possible to arrive at a test of the theory.

In 2013 Abhay Ashtekar, Ivan Agullo and William Nelson published an article in which they calculate that, under certain hypotheses, the statistical distribution of the fluctuations of this source of cosmic radiation should reveal the effect of the initial bounce: the wide-angle fluctuations should be different from those predicted by the theory that does not take quanta into account. The current state of the measurement is described in figure 9.4, where the black line represents the prediction by Ashtekar, Agullo and Nelson, and the grey dots the measured data. For now, these are not sufficient to evaluate whether the upward bend of the black line predicted by the three authors is correct or not. But measurements are getting more precise. The situation is still fluid. But those who, like myself, have spent their lives seeking to understand the secrets of quantum space are following with close attention, anxiety and hope the continuous honing of our capacity to make observations, to measure and to calculate – and are awaiting the moment in which nature will tell us whether we are right or not.